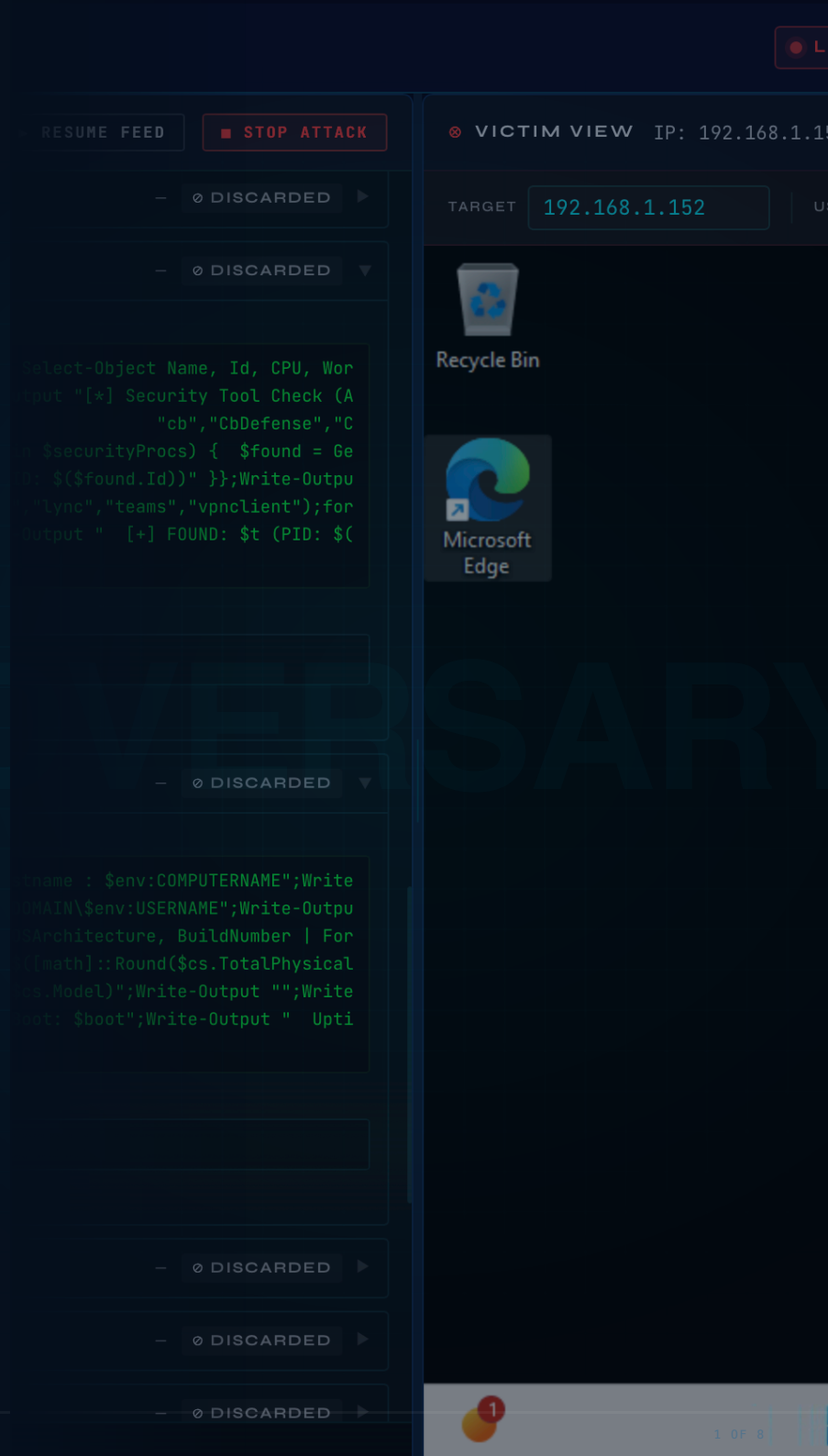
ADVERSARY SIMULATION PLATFORM

Real attacks.
Real detections.
Every time.

Automated attack scenarios, AI-generated playbooks, interactive red tools, and a fully managed Windows lab — in one integrated platform. From a customer story to a live attack to a Sophos detection in four minutes.

PREPARED BY **JASON CLARK**

ADVERSARY-SIM | LABOPS | SE-DEMO-LAB



Two Integrated Components

ADVERSARY SIMULATION

PLATFORM · ADVERSARY-SIM

24	60+	4	4
SCENARIOS	MITRE TTPS	VERTICALS	AI PROVIDERS

- **24 production attack scenarios**
  Endpoint, NDR, and Firewall coverage — each MITRE-mapped with expected detections and SE talking points
- **AI scenario generation**
  Describe an attack in plain English — the platform builds the full playbook including CALDERA abilities
- **Live split-screen attack console**
  CALDERA operation feed alongside the victim Windows desktop via Guacamole RDP — side by side
- **Browser-based red tools**
  Kali Linux, Atomic Red Team, and CALDERA — all accessible without installing a single client
- **SE talking points & scenario runbook**
  Structured pre/during/post scripts with MITRE mapping and four industry verticals per scenario
- **Scenario Studio — build or generate new attacks**
  Design scenarios manually with the MITRE technique picker, or type a plain-English description and let AI build the full playbook including CALDERA abilities

LAB_OPS

PLATFORM · LABOPS

3	2–4m	6	1
VM TYPES	PROVISION	SERVICES	COMMAND

- **One-click VM provisioning**
  Clone Windows 11, Win 11 Unmanaged, or Windows Server from Proxmox templates in 2–4 minutes
- **Start, stop, and destroy VMs**
  Full VM lifecycle from the browser — no Proxmox knowledge, no command line, no hypervisor expertise
- **Browser-based RDP via Guacamole**
  Full Windows desktop in any browser — no VPN, no RDP client, no firewall exceptions required
- **Live container health monitor**
  Status of all Docker services — Nginx, API, Guacamole, PostgreSQL, Portainer — refreshed every 15 seconds
- **Built-in guided onboarding tour**
  Contextual driver.js tour on every page — click ? anytime to walk through each feature in place
- **Quick links to Guacamole, Portainer & Proxmox**
  One-click access to the RDP gateway, container management console, and hypervisor — all from the dashboard without separate bookmarks or credentials

adversary-sim + labops = ○ se-demo-lab Deploy integrated or independently · Single Docker host per component · One `make install`

THE ATTACK FLOW — A DEMO IN SIX STEPS

1. **Choose a Scenario**
   Select from 24 production attacks across Endpoint, NDR, and Firewall. Each scenario includes expected detections, MITRE techniques, and a pre-built runbook.

2. **Target the Victim VM**
   Enter the Windows VM IP address and click Launch. The sandcat agent is pre-deployed at first boot on every LabOps VM — no manual setup needed.

3. **CALDERA Orchestrates the Attack**
   The CALDERA C2 framework dispatches abilities to the agent. The SE narrates the story — they don't operate the attack.

4. **Real Techniques Execute Live**
   Actual PowerShell, LOLBins, credential tools, and lateral movement run on the victim VM — visible in the CALDERA feed as they happen.

5. **Sophos Detects and Alerts**
   Endpoint, NDR, or Firewall generates real telemetry and alerts. The customer watches their product respond to a live attack in real time.

6. **SE Pivots to Sophos Central**
   Open the XDR threat hunt and walk the customer through the kill-chain investigation — turning a live attack into a proof of value.

11	6	7	60+	2–4m
ENDPOINT	NDR	FIREWALL	MITRE TTPS	PER DEMO

PLATFORM ARCHITECTURE — ADVERSARY-SIM

DOCKER HOST · ADVSIM-NET
- SE Console
- CALDERA C2
- n8n Workflows
- Kali Linux
- Atomic Red Team
- Guacamole RDP

↓ WINDOWS VM (LABOPS)
- sandcat agent
- Sophos Endpoint

→ SOPHOS CENTRAL
- XDR · MDR
- NDR · Firewall

SE Talking Points & Scenario Runbook

Every scenario ships with a structured runbook: what to say before the attack starts, what to highlight as techniques execute on screen, and how to pivot to Sophos Central for the detection walkthrough. Talking points are written for four industry verticals so the narrative always fits the customer in the room.

★ SE ONLY

TALKING POINTS

Discovery & Enumeration Chain

UNIVERSAL TALKING POINTS

KEY TALKING POINTS

- Every attacker runs these commands within minutes of gaining access — it's how they figure out where they are and what's valuable

LABOPS HIGHLIGHTS

VM Provisioning

Clone any Windows template from Proxmox in 2–4 minutes. Terraform handles the clone; Ansible configures RDP, creates the demo user, and disables the Windows Firewall.

2–4 MINUTES · FULLY AUTOMATED

Start · Stop · Destroy

Full VM lifecycle from the browser. Stop VMs between sessions to conserve resources. Destroy to reclaim disk after a demo. Always a confirm dialog before destruction.

GRACEFUL SHUTDOWN OR FORCE-OFF

Browser-Based RDP

Apache Guacamole renders a full Windows desktop in any browser tab. No VPN, no RDP client, no firewall exceptions. Share the URL — anyone can connect instantly.

NO CLIENT SOFTWARE REQUIRED

Service Health Monitor

Live status of all Docker containers — Nginx, API, Guacamole, guacd, PostgreSQL, Portainer — displayed as a service grid refreshed every 15 seconds.

15-SECOND REFRESH · 6 SERVICES

Terraform + Ansible

Infrastructure-as-code from start to finish. Terraform clones from Proxmox; Ansible applies day-one Windows config. Idempotent — safe to run more than once.

INFRASTRUCTURE-AS-CODE

Guided Onboarding Tour

Built-in driver.js tour covers VM lifecycle controls, provisioning flow, RDP access, and service health in context. Click ? on any page to launch the tour on demand.

ON-DEMAND · EVERY PAGE

PLATFORM ARCHITECTURE

DOCKER HOST · LABOPS-NET

- Nginx Dashboard
- LabOps API
- Guacamole
- guacd
- PostgreSQL
- Portainer

↓ TERRAFORM + PROXMOX API

PROXMOX HYPERVISOR

- VM Templates 100–199
- Terraform · Ansible

↓ CLONED FROM TEMPLATES

LAB VMS · IDS 200–299

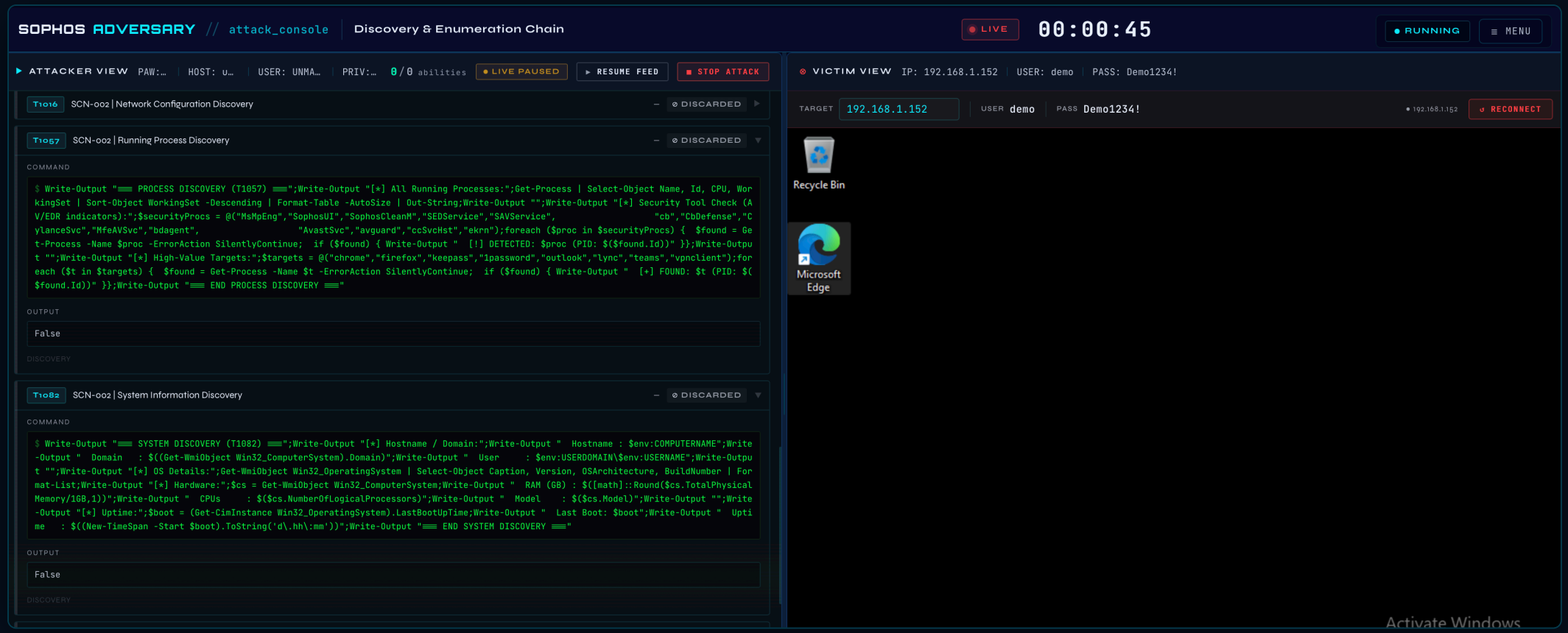
- Windows 11 + Sophos Endpoint
- Windows 11 Unmanaged
- Windows Server

↗ GUACAMOLE RDP · BROWSER-NATIVE

3	2–4m	6	1
VM TEMPLATES	PROVISION TIME	DOCKER SERVICES	INSTALL COMMAND

THE LIVE ATTACK CONSOLE

Real attack techniques executing on a real Windows endpoint — narrated in real time. The left panel streams every CALDERA ability as it runs. The right panel shows the victim desktop responding.



ATTACKER VIEW

Every CALDERA ability streams live — **technique ID, command, and output**. Narrate each step as it executes.

VICTIM VIEW

Live RDP via Guacamole — **no VPN, no client**. The customer watches their endpoint respond as detections fire.

SE NARRATION

Talking points and **expected detections** surface automatically — always know what to say next, mid-attack.

ONE CLICK TO LAUNCH

Select scenario → enter victim IP → click **Launch Simulation**. CALDERA runs the full attack chain. No CLI needed.

SCENARIO STUDIO — AI GENERATION

DESCRIBE THE ATTACK IN PLAIN ENGLISH

✦ AI GENERATE | DESIGN SCENARIO

DESCRIBE YOUR SCENARIO

A threat actor targets a mid-size manufacturing company using a spear-phishing email with a malicious Office macro. After gaining initial access, they dump credentials from LSASS memory, perform lateral movement using Pass-the-Hash to reach the domain controller, delete Volume Shadow Copies to prevent recovery, and deploy ransomware that encrypts the production floor's shared drives. The customer runs Sophos Endpoint with Intercept X.

Provider: Anthropic Claude

GENERATE SCENARIO →

WHAT AI GENERATES IN UNDER 60 SECONDS

- **MITRE ATT&CK mapping** — technique IDs and tactic names auto-assigned from the description
- **CALDERA adversary profile** — abilities and PowerShell commands loaded directly into the C2 framework
- **Expected Sophos detections** — severity-graded, product-specific, ready for the demo narrative
- **Customer talking points** — four industry verticals: Universal, Financial Services, Healthcare, Manufacturing
- **SE runbook** — setup steps, timing notes, and pivot instructions for Sophos Central

Immediately Launchable

Generated scenarios load directly into CALDERA — no manual ability creation, no adversary configuration. The moment generation completes, the scenario appears in the SE Console ready to launch.

AI PROVIDERS

Anthropic Claude | OpenAI GPT-4o | Google Gemini | Ollama (local)

From a customer story to a live attack to a Sophos detection — four minutes.

This is the demo that wins technical buyers.

PREPARED BY JASON CLARK

This platform replaces canned demos, screen recordings, and verbal descriptions of what Sophos detects. Every demo runs live, against a real Windows VM, with real attack techniques executing in real time. The customer doesn't watch a video — they watch their product respond to an actual attack.

VALIDATES THE SOPHOS PORTFOLIO

ENDPOINT · NDR · FIREWALL · XDR · MDR · CENTRAL · EMAIL

SOPHOS ADVERSARY SIMULATION

SCENARIOS	24 production · 68 total incl. AI-generated
PRODUCTS	Endpoint · NDR · Firewall
MITRE TTPS	60+ techniques mapped
DEMO DURATION	2–4 minutes per scenario
TALKING POINTS	4 industry verticals / scenario
AI PROVIDERS	Anthropic · OpenAI · Gemini · Ollama
C2 FRAMEWORK	CALDERA + sandcat agent
RED TOOLS	Kali Linux · Atomic Red Team
DEPLOYMENT	Docker · `make install`

SOPHOS LAB_OPS PLATFORM

VM TEMPLATES	Win 11 · Win 11 Unmanaged · Server
VM ID RANGE	200–299 (templates 100–199)
PROVISIONING	Terraform + Proxmox API · 2–4 min
VM CONFIG	Ansible – RDP, demo user, firewall
RDP ACCESS	Apache Guacamole · browser-native
SERVICES	Nginx · API · Guacamole · Portainer
HYPERVISOR	Proxmox VE
NETWORK	labops-net 172.20.0.0/24
DEPLOYMENT	Docker · `make install`

Get Access

Repo · **github.com/sophos-internal/se-demo-lab**

Install guide in README · One `make install` per component

FULL PLATFORM – SE-DEMO-LAB

adversary-sim and labops share a Docker bridge network (`labops-net`) and deploy on a single host. Install independently with `make install`, or together as se-demo-lab. Extensible: any additional demo stack joins by declaring `labops-net` as an external network.

Built-in help.
No ticket required.

200 SEs in 30 countries running live attack demos. When something goes wrong before a customer call, there's no time to file a support request. Every page in the platform ships with a built-in troubleshooting guide — searchable, offline, and honest about its own limitations.

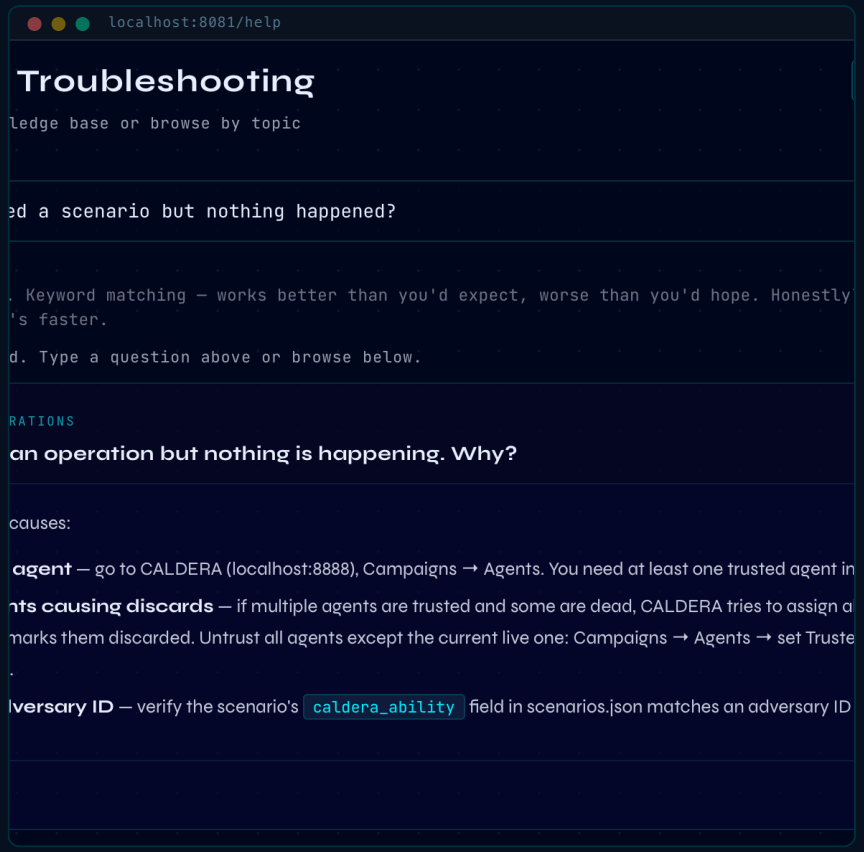
Keyword Search
Type what's broken. Synonym expansion matches "sandcat" to "agent", "RDP" to "Guacamole", "restart" to "reset". Top 3 answers ranked by relevance.

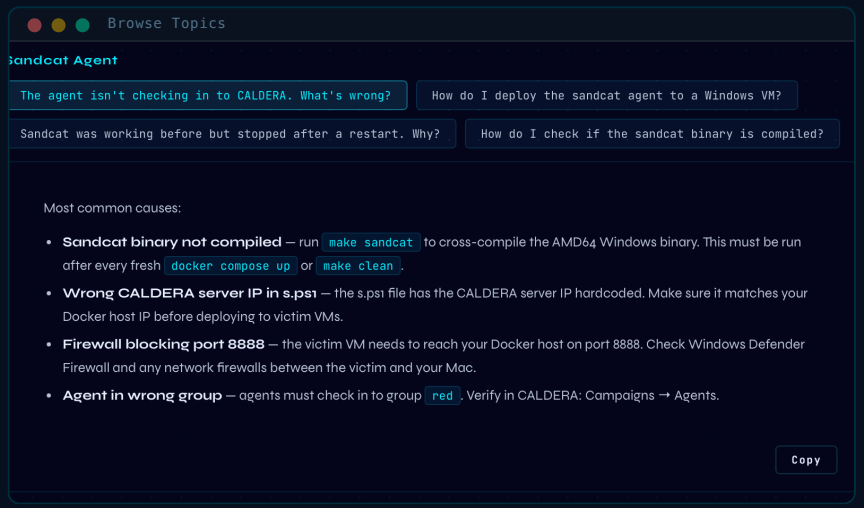
38 Answers, 9 Topic Categories
Getting started, sandcat agent, CALDERA operations, RDP/Guacamole, AI providers, VM management, Docker infrastructure, credentials — every common failure mode covered.

Zero Dependencies
Pure client-side keyword matching. No LLM calls, no API keys, no hallucinations. Works offline. The KB file is a plain markdown file — any SE can read, edit, and update it directly.

Contextual Entry Points
Error states in the SE Console surface a direct link to the relevant KB section. The `?` tour icon on every page opens guided walkthroughs before reaching the full help center.

Experimental keyword matching — works better than you'd expect, worse than you'd hope. Honestly? Browse the FAQ — it's faster.





HELP CENTER · LOCALHOST:8081/HELP